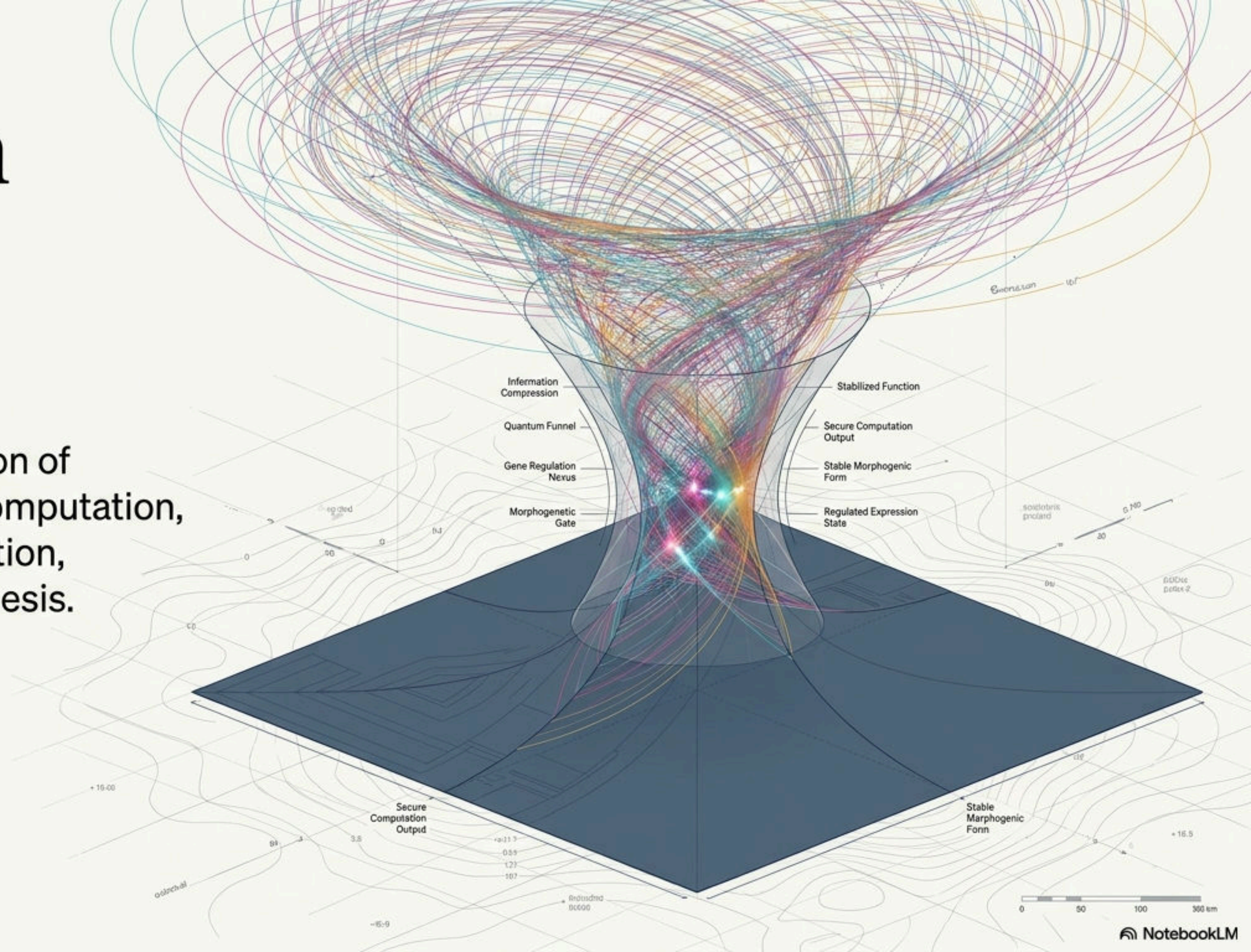


Function Survives Collapse

A geometric unification of privacy-preserving computation, bacterial gene regulation, and plant morphogenesis.

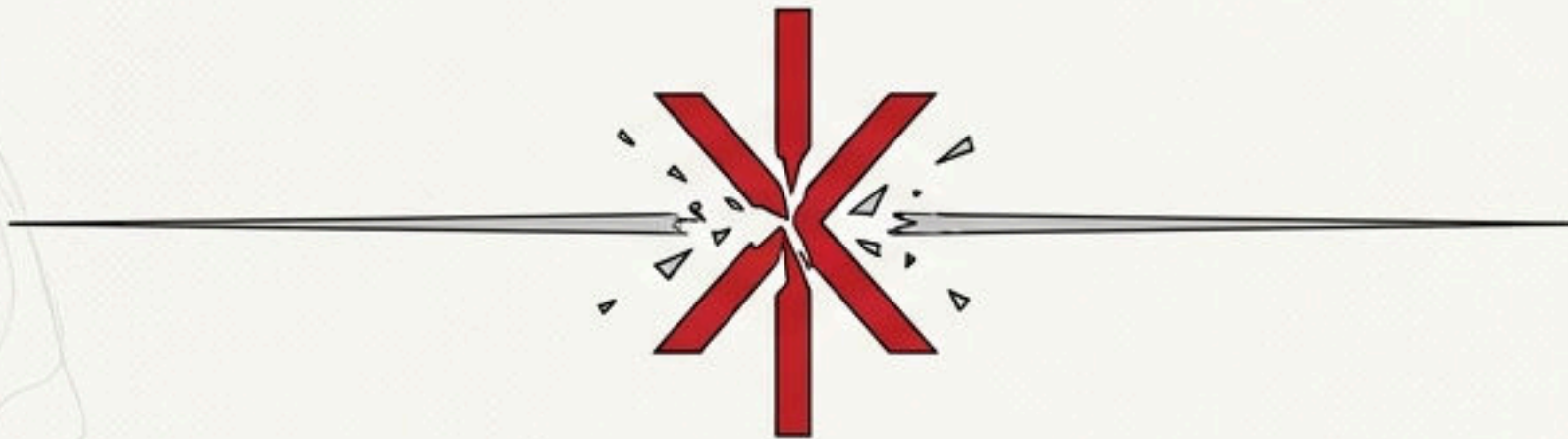


The linear pipeline ontology fails to describe robust systems

Historically, computation and biology were modeled as pipelines: sequences of information passed along brittle channels. The reality is a geometry of projection, where functional output is stabilized precisely by sacrificing the path used to reach it.

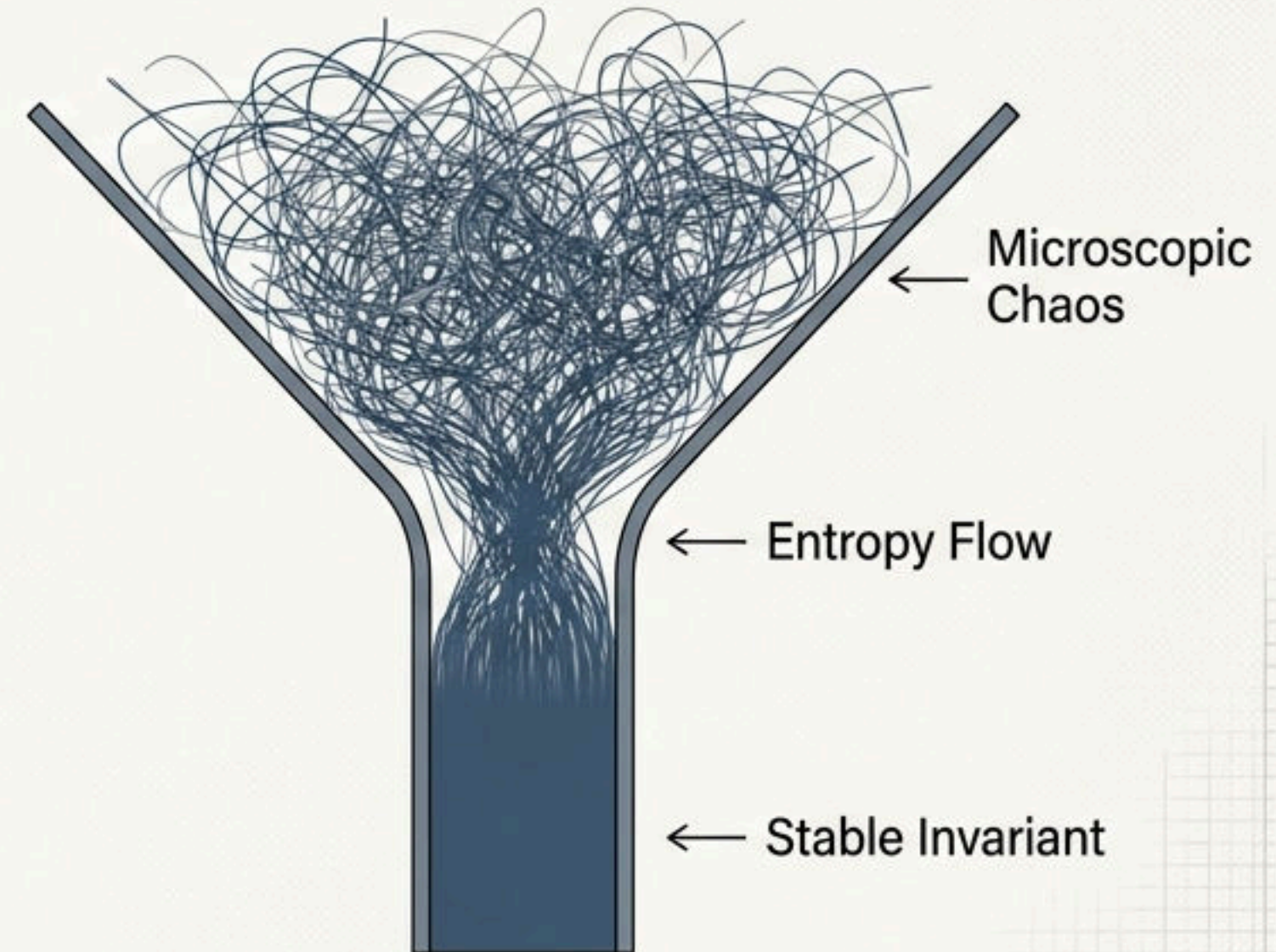
The Pipeline

Sender → Channel → Receiver



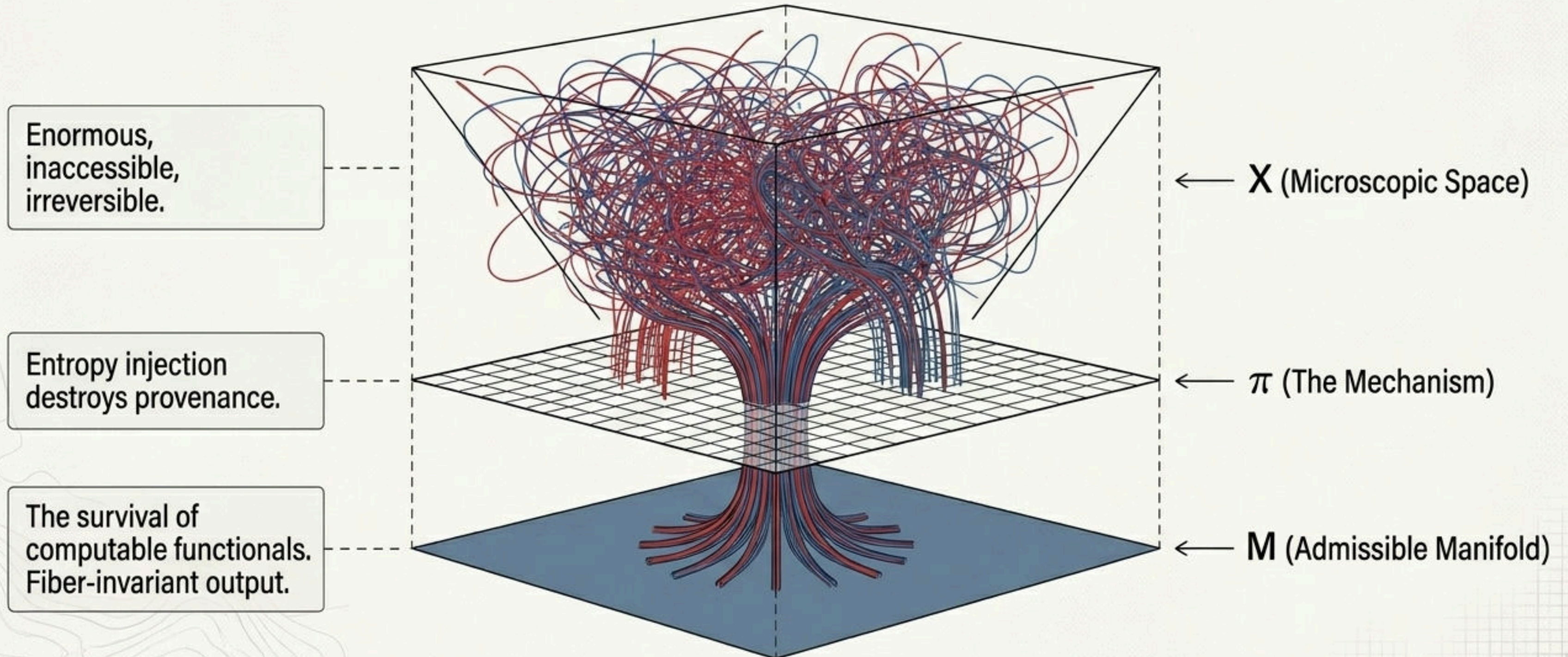
DNA → RNA → Protein

The Projection



Admissible Projection maps microscopic chaos to functional invariants

Robust systems operate via an irreversible projection map. They do not filter noise to preserve a signal; they consume the microscopic trajectory itself to fuel the survival of a low-dimensional functional invariant.



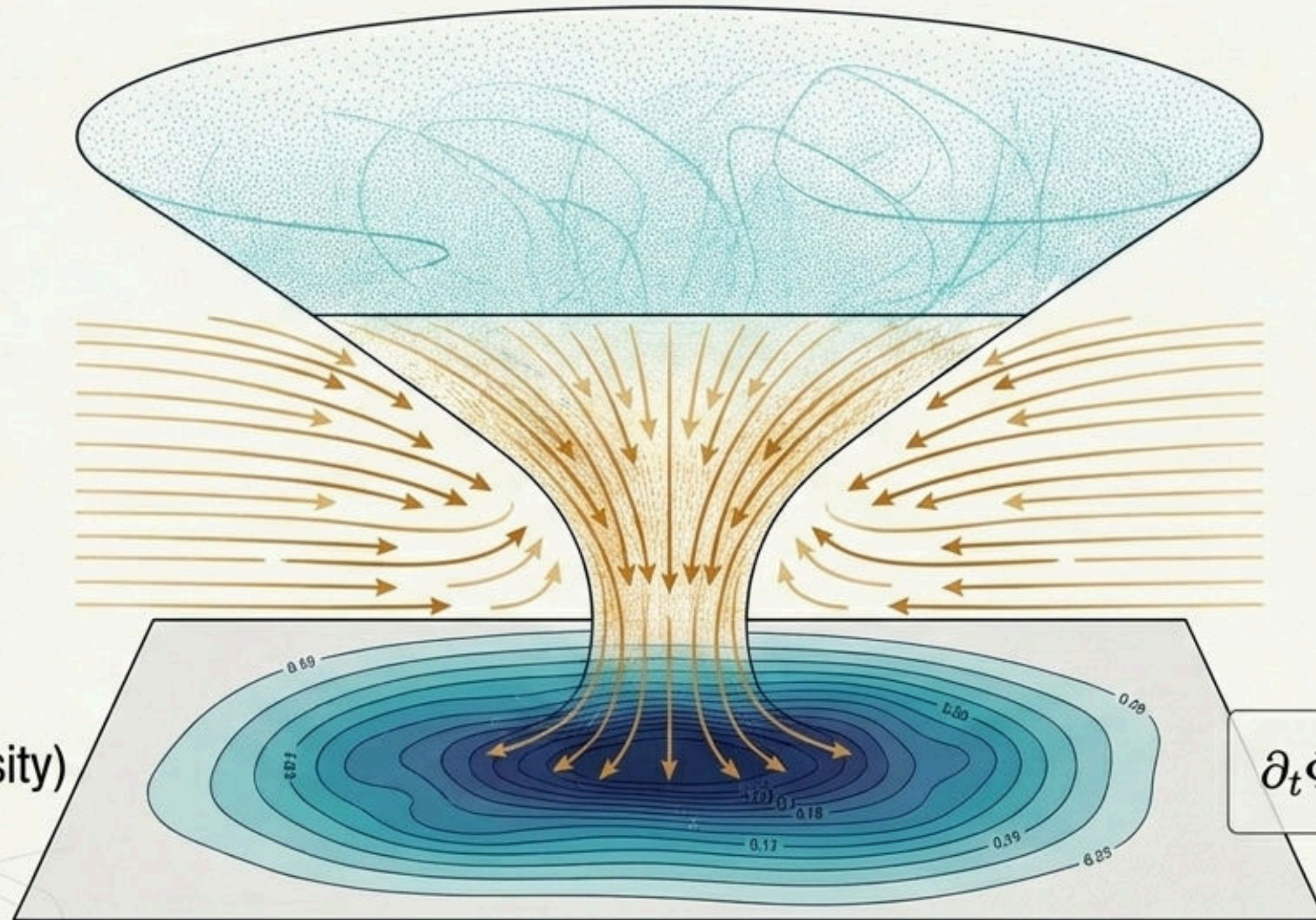
The RSVP field triple governs the dynamics of invariant persistence

Admissible projection is the mathematical attractor of constrained Euler-Lagrange flow. Systems navigate topology through admissibility density, trajectory flow, and geometric entropy.

S (Entropy Field)

v (Trajectory Flow)

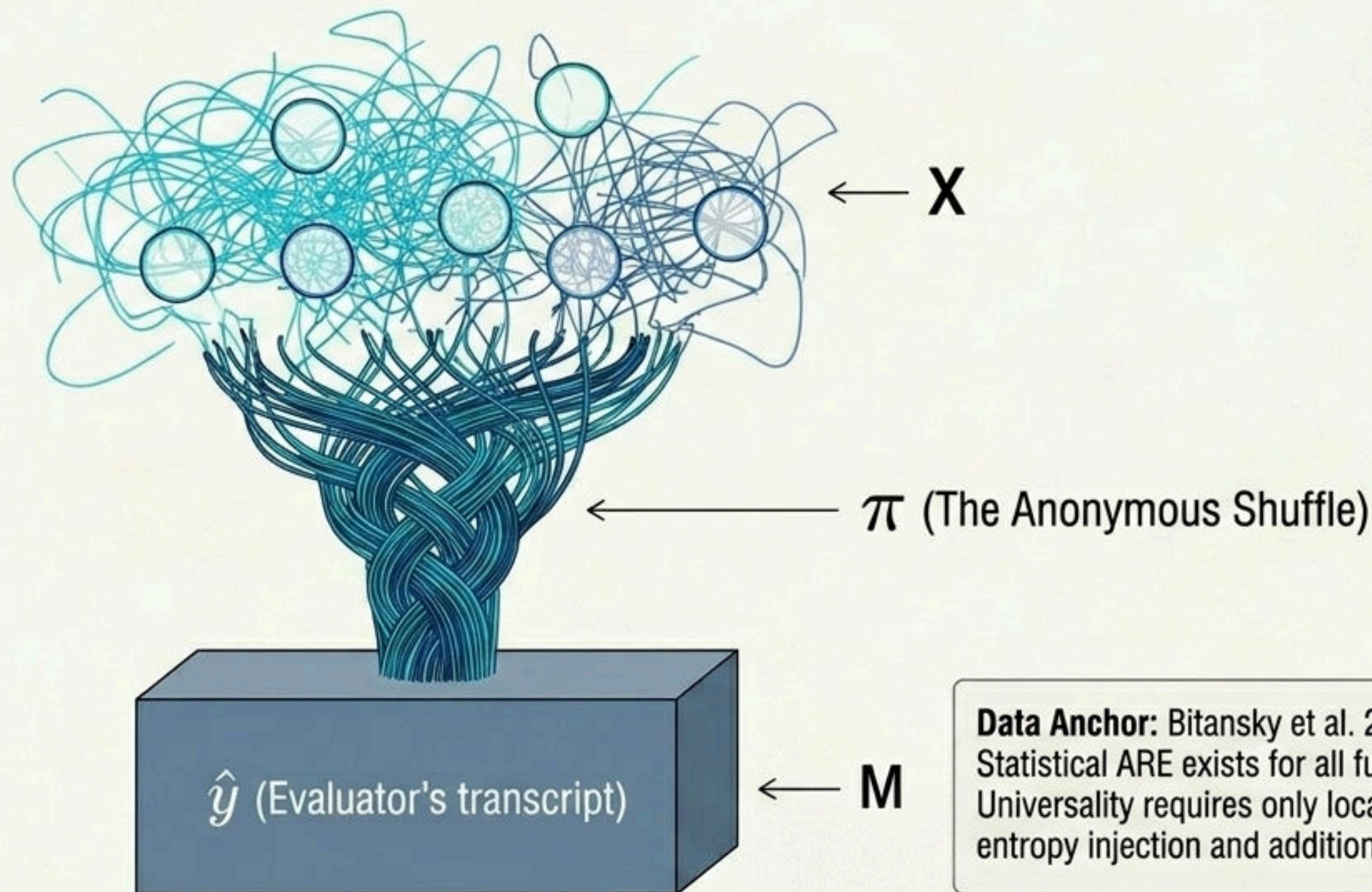
Φ (Admissibility Density)



$$\partial_t \Phi + \nabla \cdot (\Phi \mathbf{v}) = -\lambda S$$

Cryptography achieves universal computation through entropy mixing

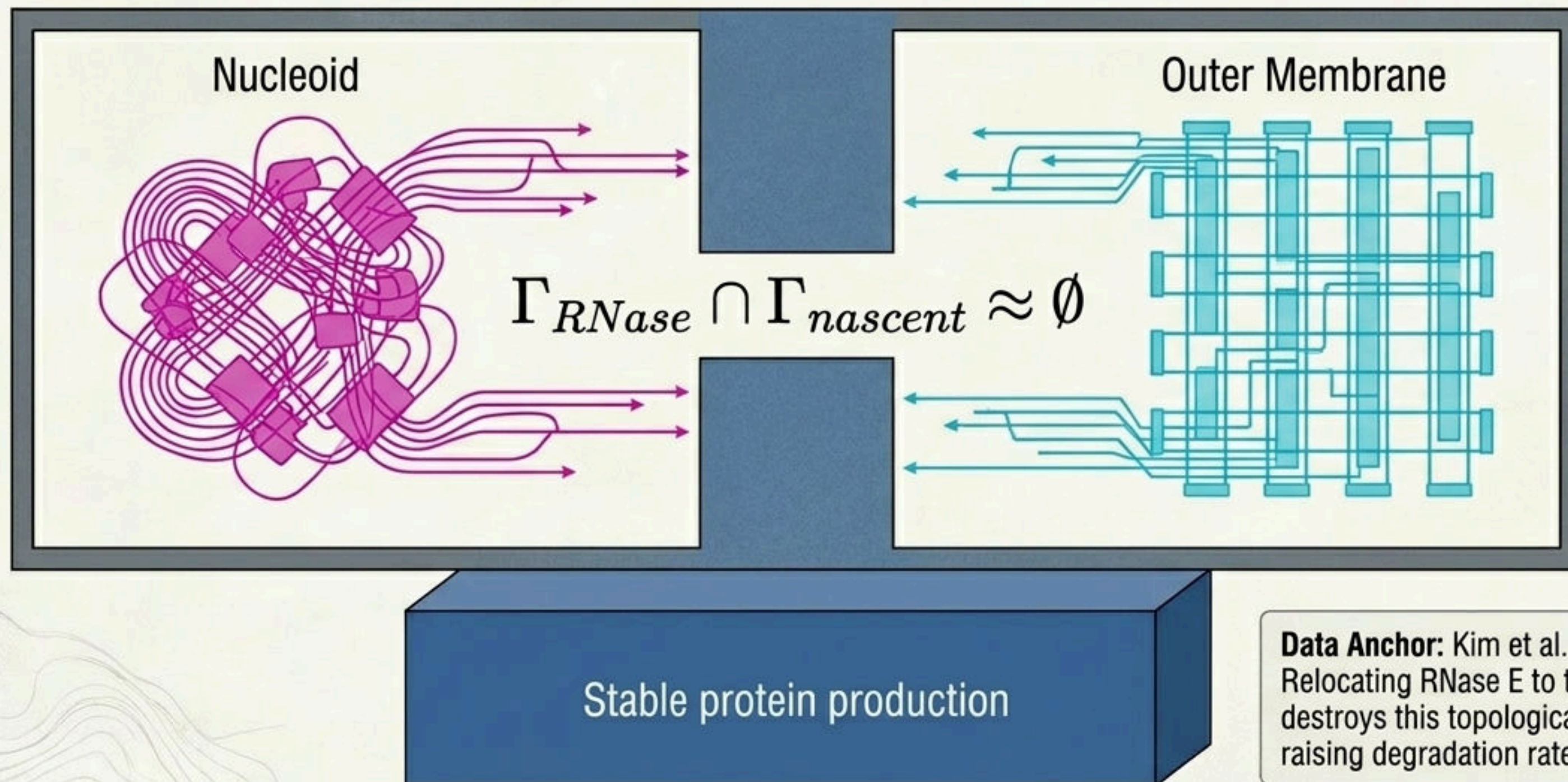
In Additive Randomized Encodings (ARE), secure computation does not require a pipeline. Local randomness thermalizes the microscopic state. Provenance is irreversibly destroyed by the shuffle.



Data Anchor: Bitansky et al. 2025.
Statistical ARE exists for all functions.
Universality requires only local
entropy injection and addition.

Bacterial gene expression preserves function through spatial separation

Transcription and translation are not a universal pipeline. Nascent transcripts survive because degradation machinery is physically anchored, creating a topological obstruction.

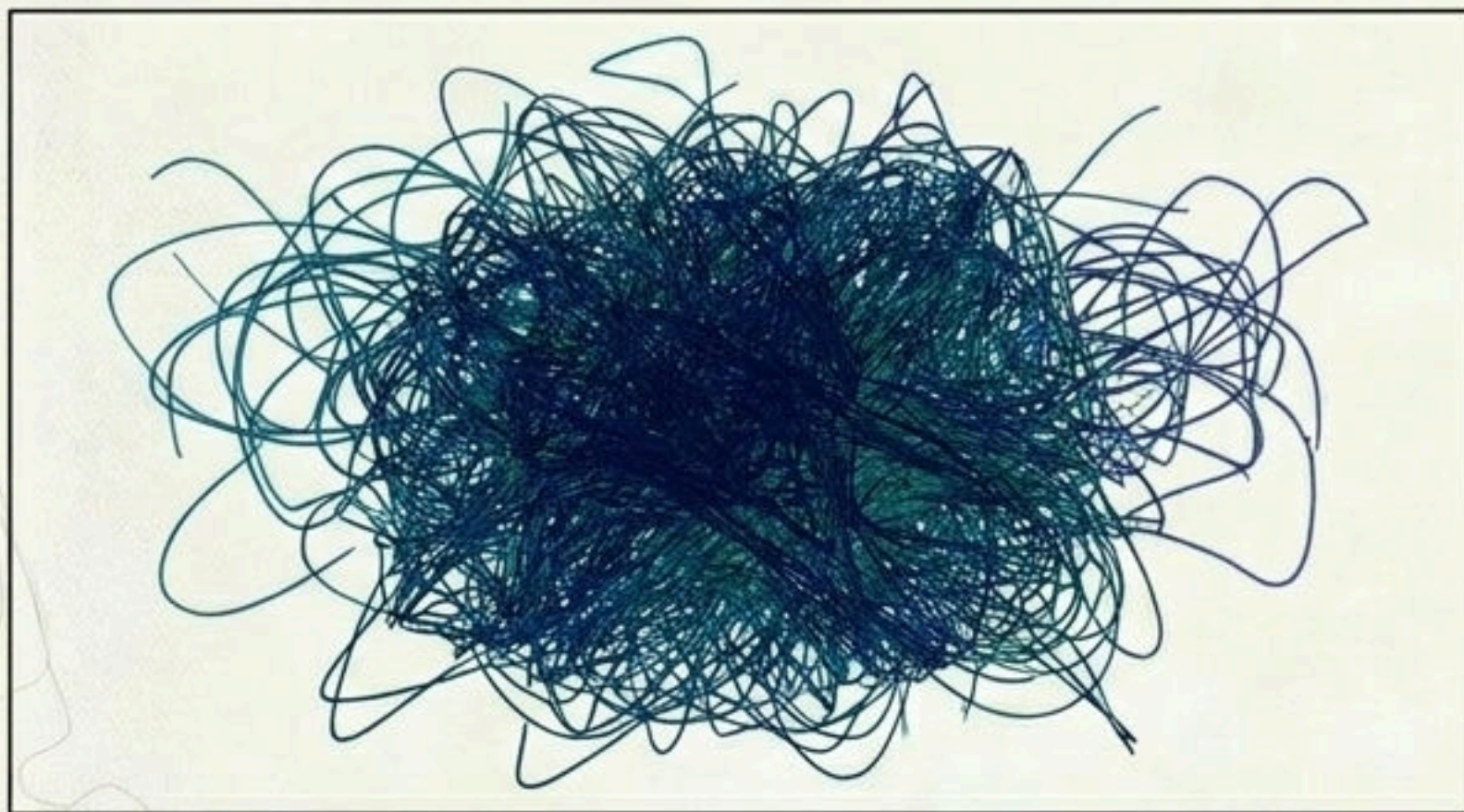


Data Anchor: Kim et al. 2026.
Relocating RNase E to the cytoplasm destroys this topological barrier, raising degradation rates by 7x.

Mixing and separation are geometric duals of topological privacy

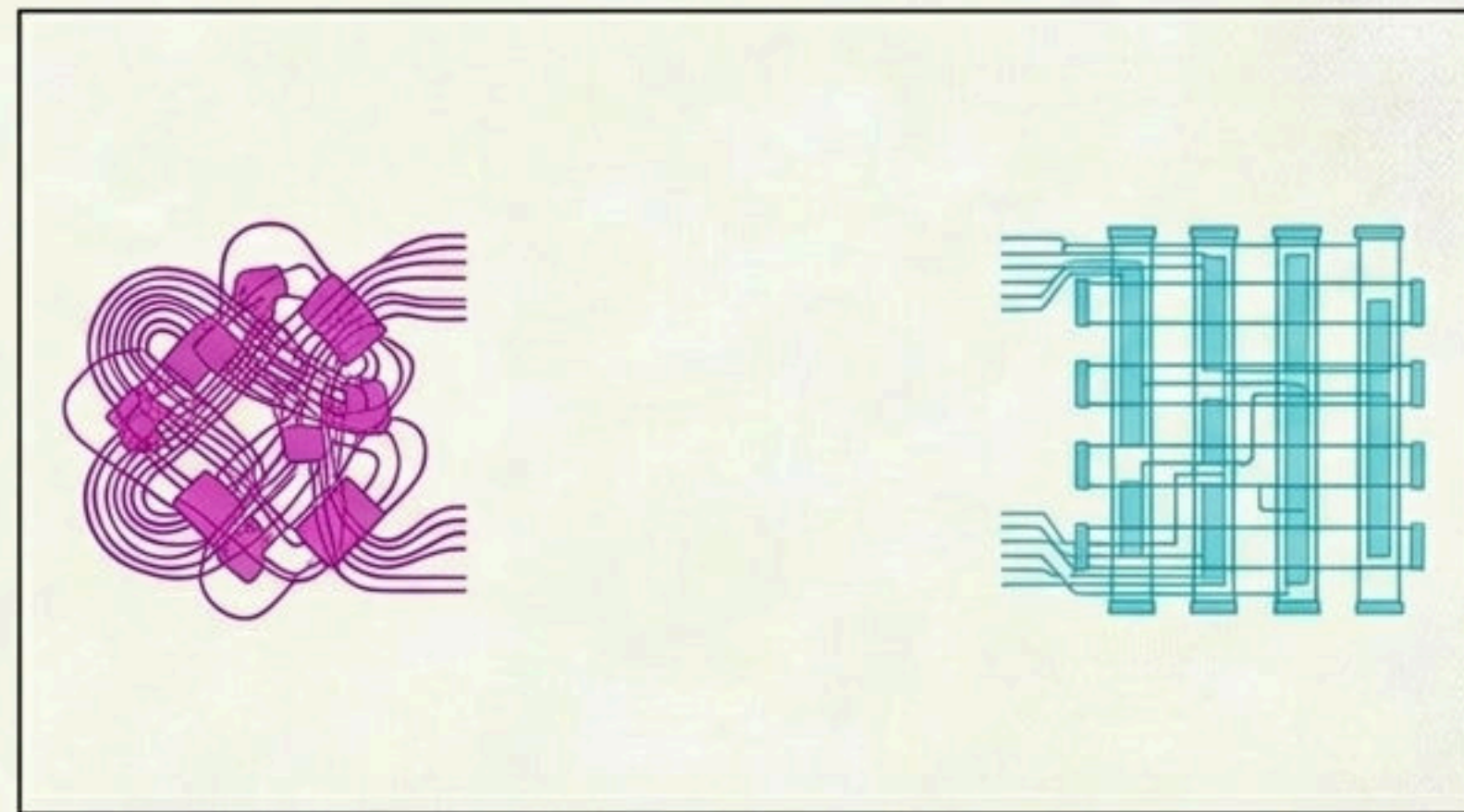
Both crypto and biology protect their functional invariants by denying observers access to the microscopic interior, achieving identical unrecoverable provenance.

Mixing (Cryptography)



Cohomological trivialization. The interior is crowded with indistinguishable alternatives. Extension is impossible because there are too many paths.

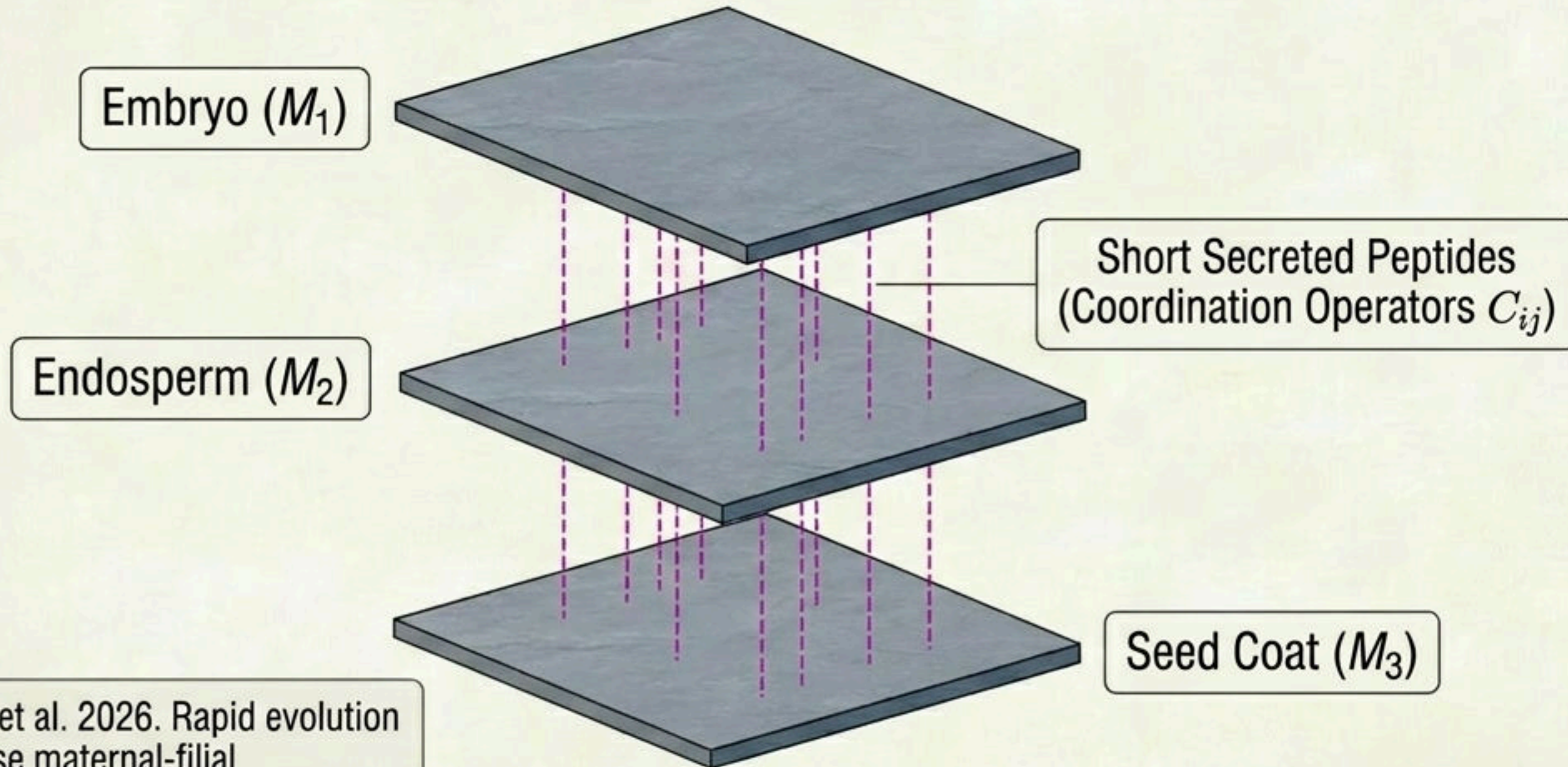
Separation (Biology)



Cohomological obstruction. The interior is empty of reachable paths. Extension is impossible because the geometry forbids the encounter.

Developing seeds propagate constraints across isolated manifolds

Global developmental coherence emerges from sparse, low-dimensional signaling projections across symplastically isolated tissue interfaces.



Data Anchor: Martin et al. 2026. Rapid evolution occurs exactly at these maternal-filial interfaces to maintain coordination while allowing internal tissue trajectories to vary.

The Isomorphism Matrix confirms structural equivalence across domains

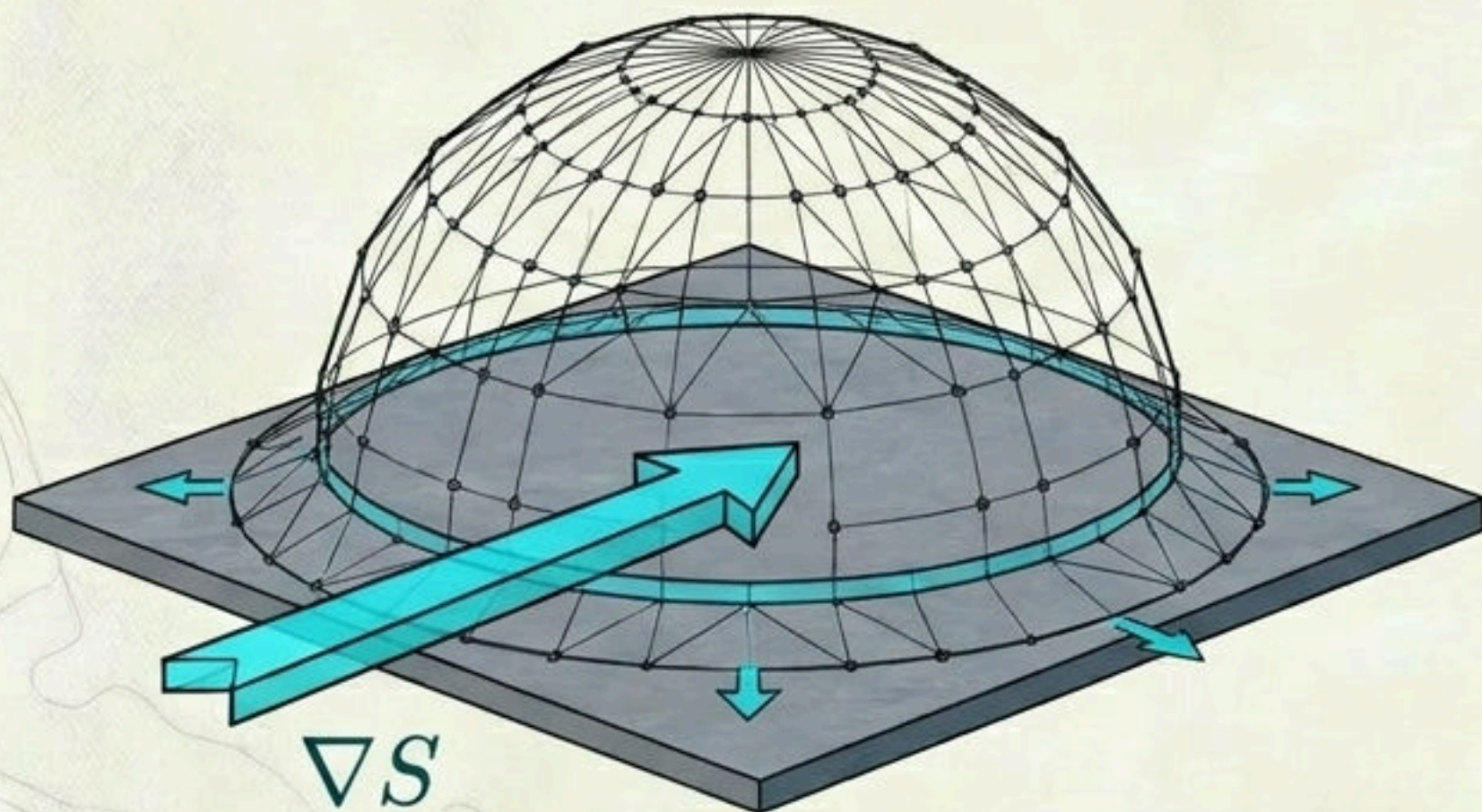
Despite radically different substrates, all four domains map 1:1 onto the precise mathematical architecture of Admissible Projection.

	Cryptography	Bacterial Regulation	Seed Development	Typography
Microscopic Space (X)	Joint Input Space	Molecular Trajectories	Multi-Tissue Biochemistry	Stroke/Pixel Geometry
Projection Mechanism (π)	Anonymous Shuffle	Spatial Obstruction	Interface Signaling	Perceptual Recognition
Entropy Injector (S)	Local Encoders	Spacer RNA/Diffusion	Genetic Conflict	Stylistic Variation
Functional Invariant (M)	Secure Output Function	Protein Abundance	Viable Seed Maturation	Semantic Glyph Identity
Failure Mode	Decoding Error	Cytoplasmic RNase E	Morphological Defect	Illegibility

Function destabilizes only when entropy turns transverse

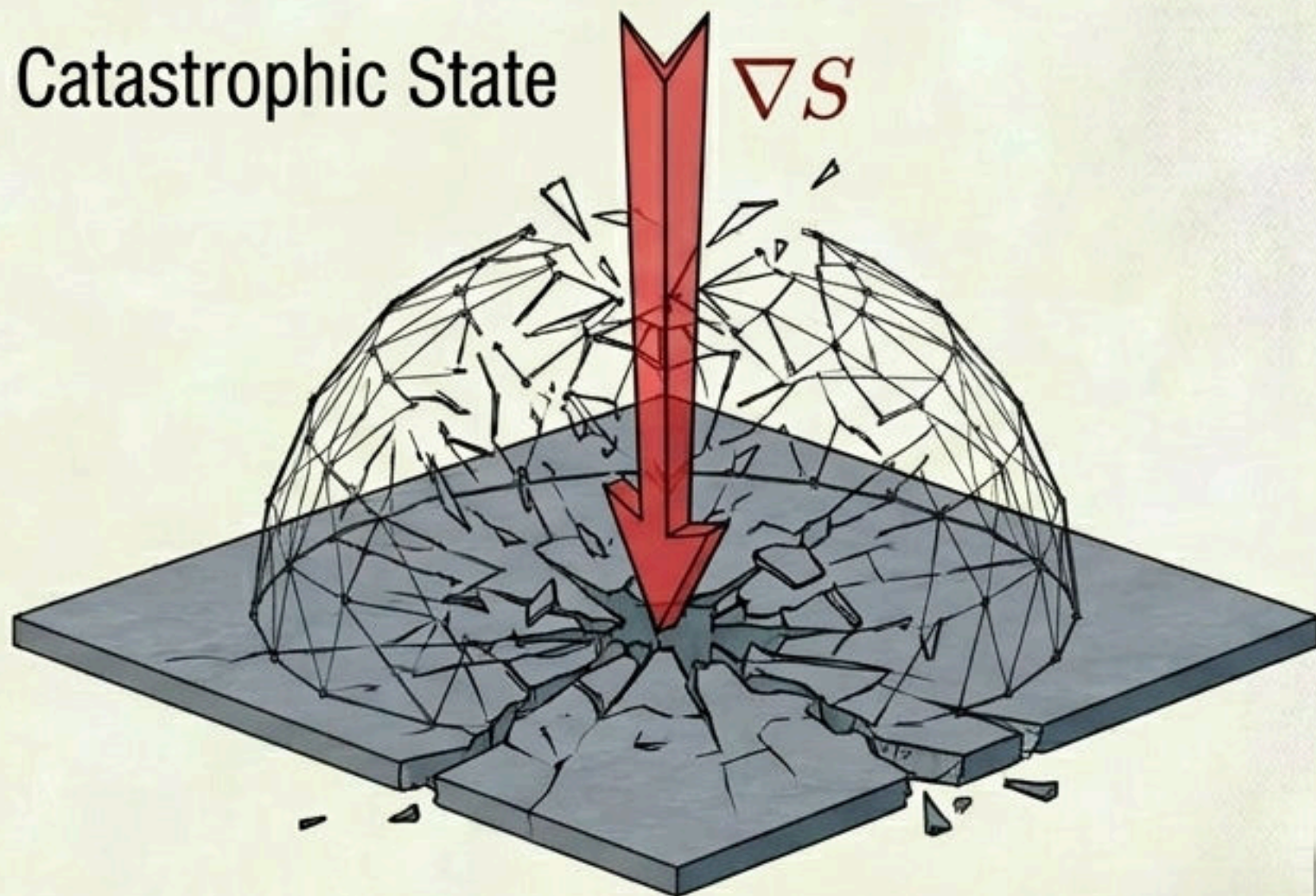
A robust system absorbs entropy to enlarge its invariant fibers. Catastrophic collapse occurs only when entropy flow breaks the tangency of the Admissibility Bubble.

Safe State



Robustness increases without leaving the manifold.

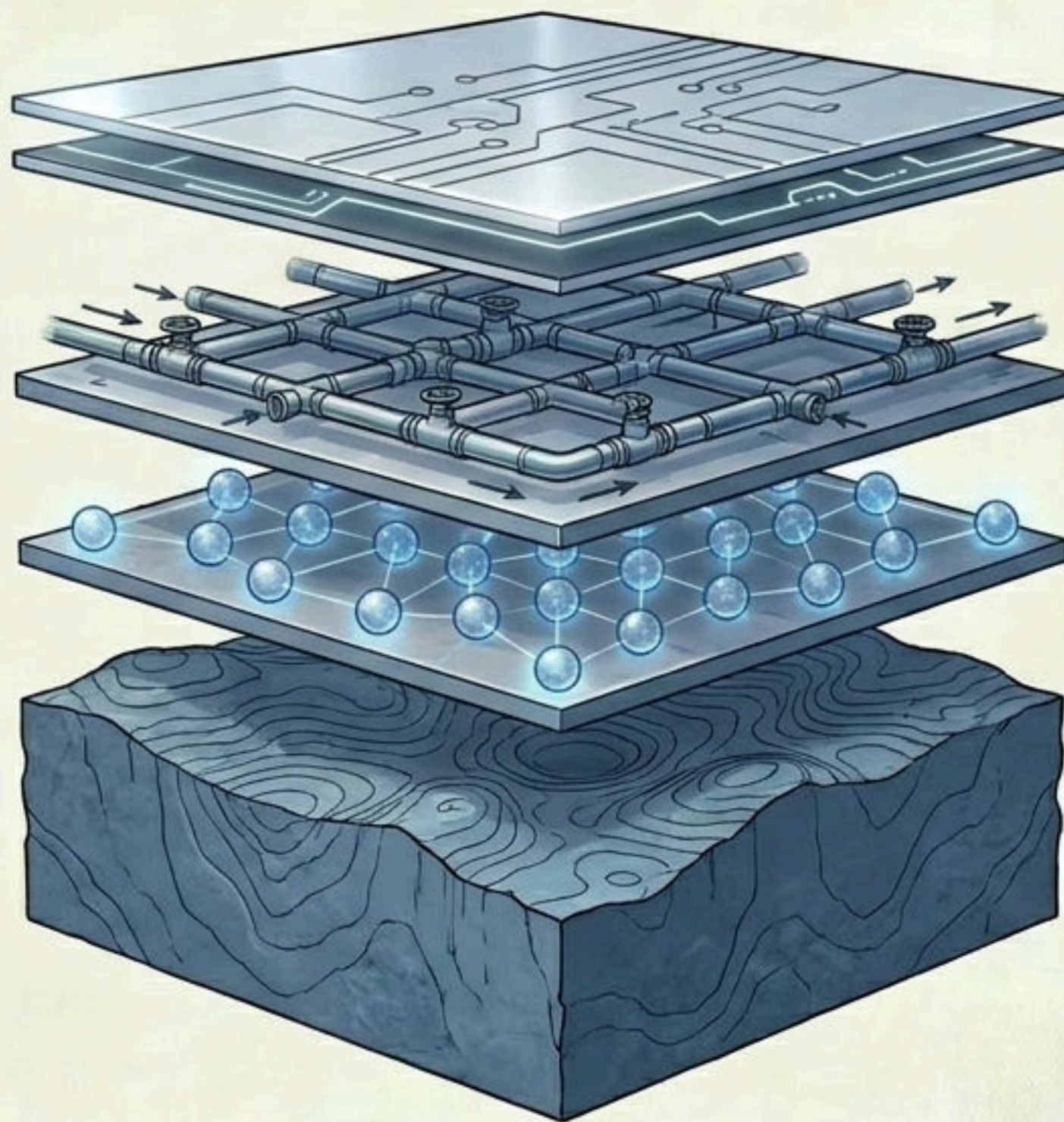
Catastrophic State



Transverse flow breaks the manifold. Projection fails.

The architecture of admissible systems operates across four layers

To unify continuous physics with discrete topologies, the framework separates the dynamical flows from the irreversible collapse events.



Layer 4: Adm (Categorical Abstraction).
Morphisms, observable algebras, and domain-independent structural equivalences.

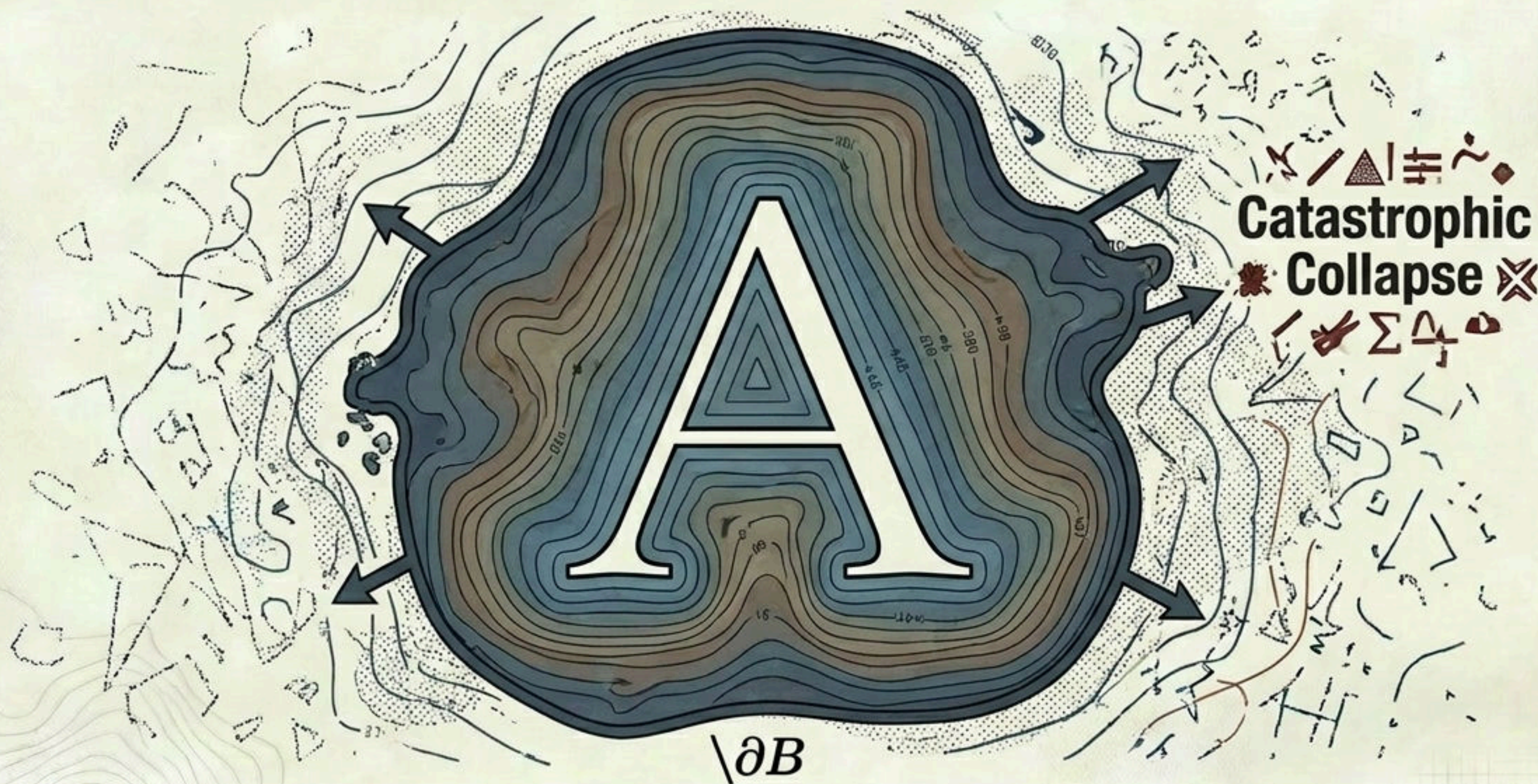
Layer 3: Yarncrawler (Network Transport).
Inter-bubble transport routing constraints across distinct admissibility basins.

Layer 2: Spherepop (Discrete Topology).
Discrete irreversible operators crossing the bubble boundary.

Layer 1: RSVP (Continuous Physics).
The foundational differential geometry of the field triple.

Typography visually instantiates admissible projection geometry

A glyph is not a bitmap; it is a projection class. The letter 'A' represents a vast microscopic trajectory space of strokes mapping onto a single, stable semantic invariant manifold.

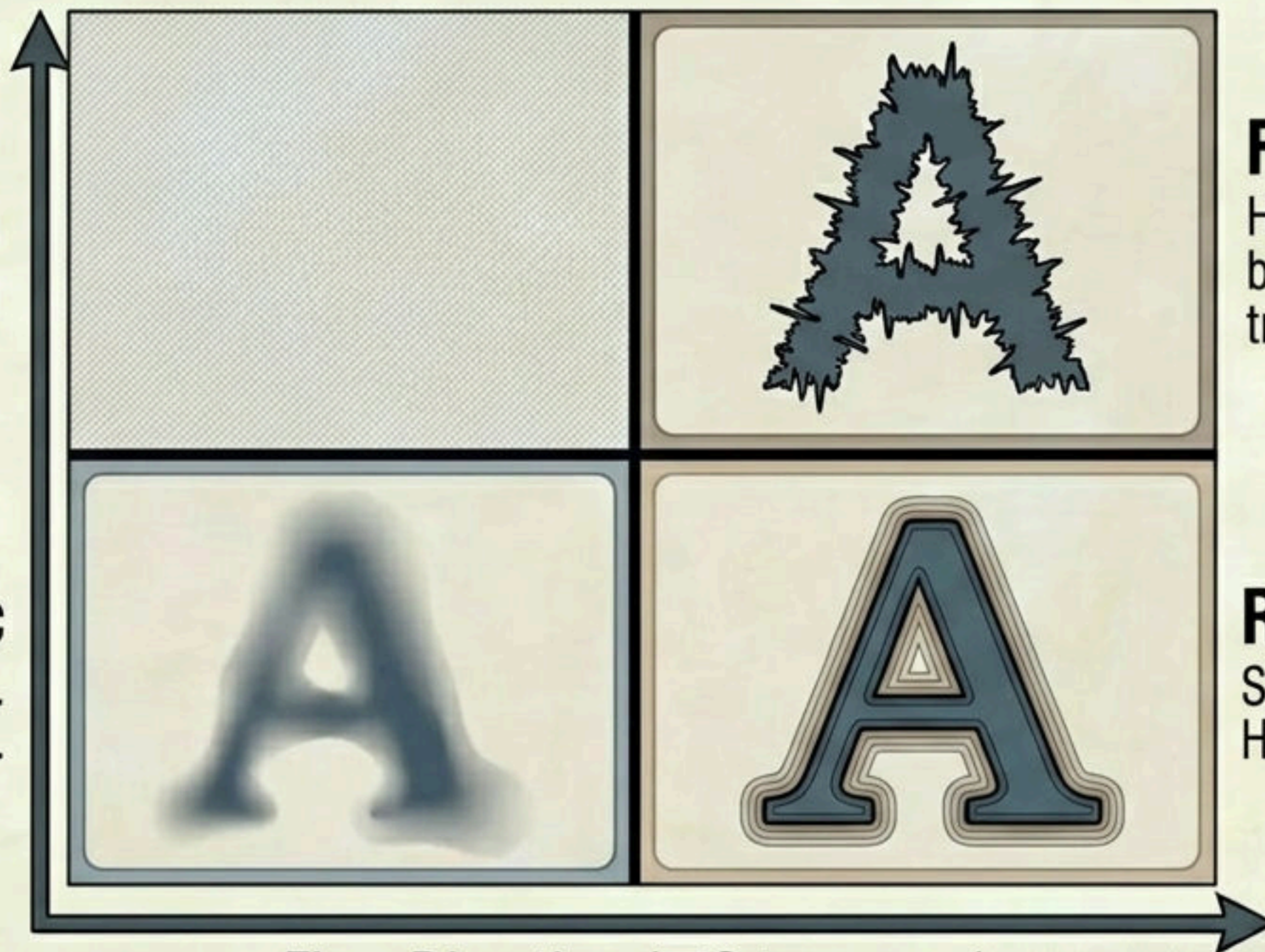


Font styles are distinct thermodynamic flows within the trajectory space

Frenetic styles inject high entropy but maintain semantic stability by keeping that entropy tangential to the manifold.

Entropy Level (S)
(Low to High)

Hypnagogic
Slow boundary oscillation.
Dreamlike ambiguity.



Frenetic

High microscopic disorder,
but readable because
transverse flow is zero.

Regular / Bold

Stroke mass expands uniformly.
High recognizability density.

Flow Direction (∇S tangency)
(Transverse to Tangent)

Reading is a dynamical attractor recovering the projection class

Adversarial typography occurs when sensory inputs sit exactly on the boundary between overlapping admissibility bubbles, destabilizing the RSVP flow.

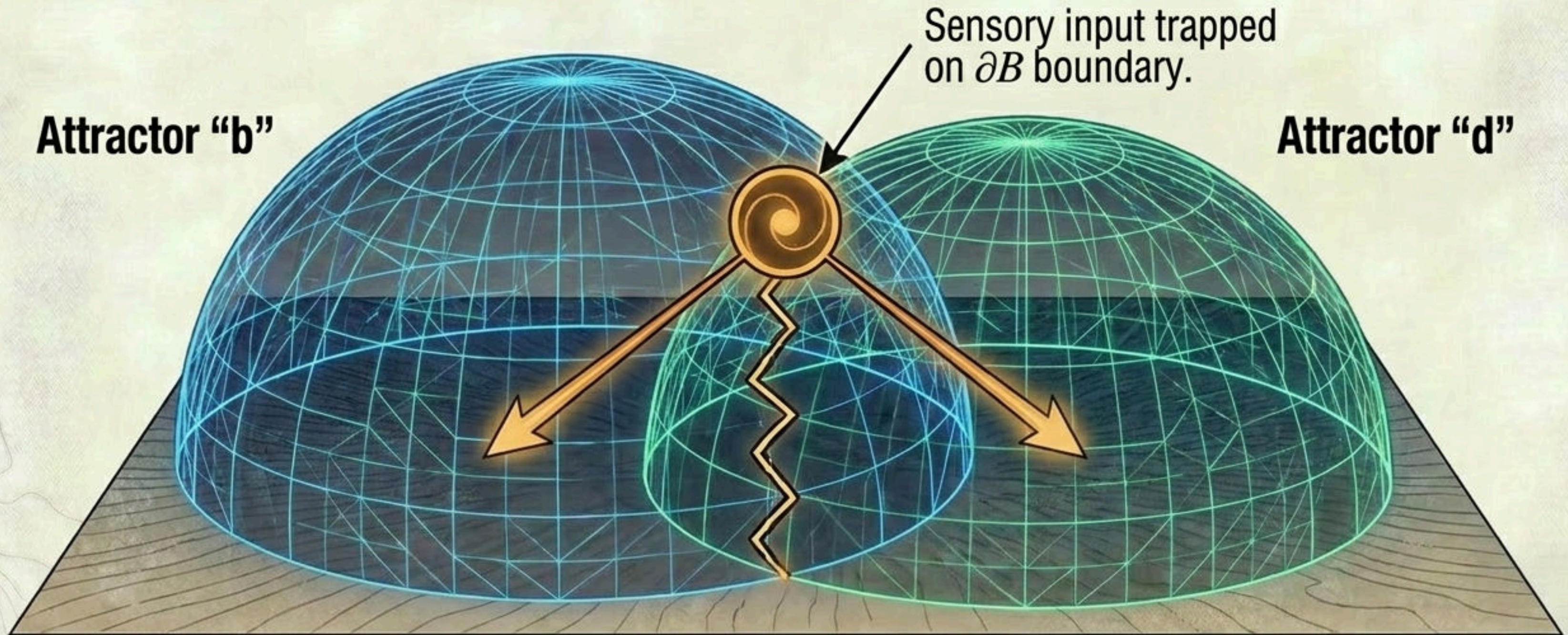
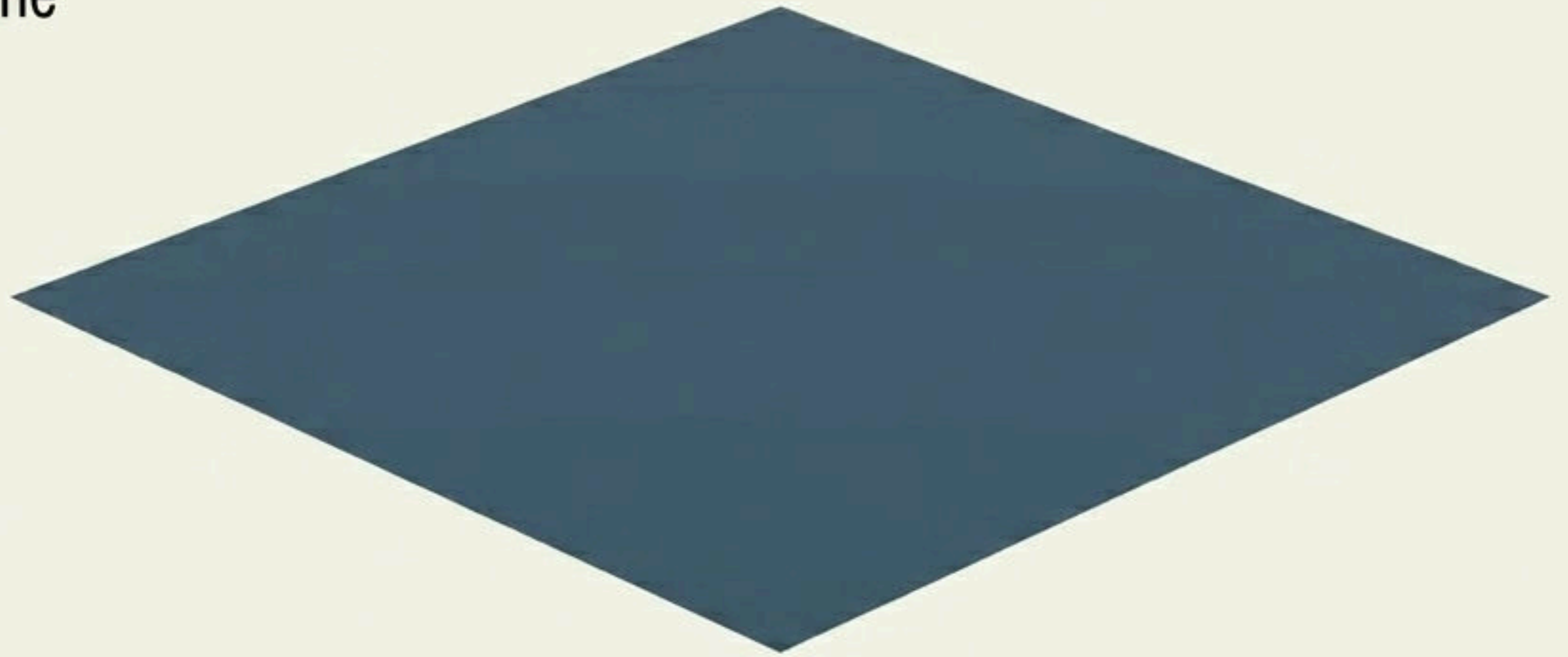


Figure 3: Topological Boundary Instability in Adversarial Typography.

The map is not the territory; the map is the attractor

The exact microscopic trajectory is irretrievable. The pipeline breaks. The invariant holds. Stability belongs exclusively to the projection class.



What survives collapse.